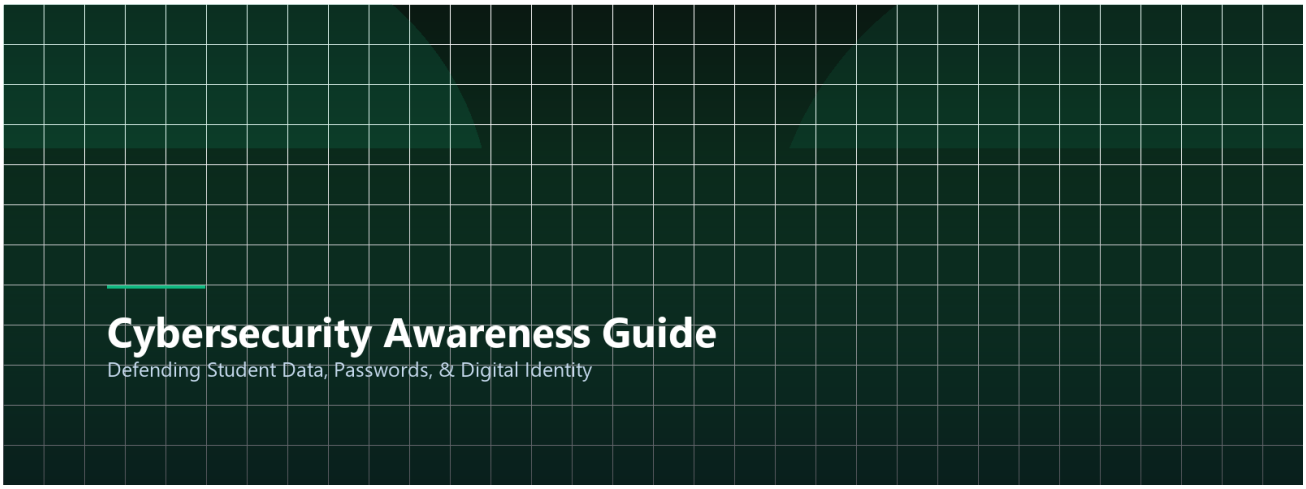

Cybersecurity Awareness Guide for Students

Defending Student Accounts, Credentials, and Personal Data Against Digital Threats

Course: Information Security & Platform Technology (COSC1008)
Institution: STI West Negros University, Bacolod City
Author: Jhon Paulo Gascon (2nd Year BSIT)
Scope: Phishing Heuristics, Multi-Factor Authentication, & Public Wi-Fi Security

1. Phishing Recognition & Social Engineering

Phishing represents over 80% of reported student cybersecurity compromises. Adversaries fabricate emails impersonating university registrars, LMS platforms, or mobile banking apps, coercing victims into entering login credentials under false pretexts of urgent account suspension.

> **KEY CONCEPT — Social Engineering:** The psychological manipulation of individuals into voluntarily disclosing confidential security information or executing malicious software.

Anatomy of an Attack: The 4 Critical Red Flags

Indicator	Deceptive Tactic	Verification Protocol
Artificial Urgency	'Account will be disabled in 24 hours'	Institutions never demand instant credential verification
Lookalike Domain	support@sti-wnu-verify.com instead of .edu.ph	Inspect full sender header before clicking links
Mismatched Hyperlinks	Hovered destination leads to external IP	Inspect URL preview on desktop; avoid tapping blind on mob
Suspicious Attachments	grades_midterm.zip or .exe executable files	Never download unverified archives from outside portal

2. Password Entropy & Multi-Factor Authentication

Modern automated brute-force tools running on consumer GPUs can test billions of password combinations per second. Short complex passwords (e.g., 'P@ss12!') are cracked in minutes, whereas long passphrases ('coffe-book-bacolod-sky') require centuries to compute.

Character Length	Complexity Composition	Brute-Force Crack Time	Security Rating
8 Characters	Numbers + Mixed Letters	~ 1 Hour	Vulnerable (Unacceptable)
10 Characters	Letters, Numbers, Symbols	~ 3 Weeks	Marginal
14 Characters	Passphrase (4 random words)	~ 2,000 Centuries	Strong (Recommended)
16+ Characters	Managed via Password Vault	Trillions of Years	Bank-Grade

KEY CONCEPT — Multi-Factor Authentication (MFA): MFA mandates two separate verification proofs: Something you know (password) + Something you have (authenticator app OTP). This stops 99.9% of automated credential stuffing attacks.

3. Public Wi-Fi Defense & Session Security

Connecting to unsecured public Wi-Fi in cafes and commuter lounges exposes unencrypted student traffic to Man-in-the-Middle (MitM) eavesdropping and packet sniffing. Malicious actors deploy rogue access points mimicking legitimate campus SSIDs.

Security Protocol	Vulnerability Risk	Mandatory Defensive Action
Open Unencrypted Wi-Fi	Packet sniffing & cleartext capture	Never login to financial or portal accounts without a trusted VPN
HTTPS Encryption	DNS spoofing / SSL stripping	Always check browser address bar for valid TLS certificate lock
AirDrop / Nearby Share	Unsolicited malware broadcast	Set mobile visibility to 'Contacts Only' or disable while traveling
Auto-Connect SSIDs	Rogue hotspot impersonation	Disable 'Connect Automatically to Open Networks' in OS settings

4. Quick Review Questions & Source Credits

Test your grasp of core information security principles:

1. **What is the primary indicator distinguishing a legitimate university URL from a phishing lookalike domain?**
2. **Why does a 14-character passphrase offer exponentially higher brute-force resistance than an 8-character complex pas**
3. **Define Multi-Factor Authentication (MFA) and explain why SMS-based verification is vulnerable to SIM-swap attacks.**
4. **What security hazard is introduced when a student laptop auto-connects to unverified open Wi-Fi networks?**
5. **Under the Cybercrime Prevention Act of 2012 (RA 10175), what legal consequence is prescribed for unauthorized syste**

Image & Visual Credits

- *Cover Banner: Cybersecurity Awareness Shield — Original high-resolution render generated via Antigravity visual engine.*
- *Entropy Benchmark Data: NIST Special Publication 800-63B Digital Identity Guidelines.*
- *Statutory Citations: Republic Act No. 10175 (Cybercrime Prevention Act of 2012), Republic of the Philippines.*